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Accidental hypothermia in adults: Clinical manifestations and evaluation

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INTRODUCTION

Death from exposure and accidental hypothermia occurs throughout the world and can present significant management problems [1-4]. While typically associated with regions of the world with severe winters, hypothermia is also seen in areas with milder climates, such as the southern United States [5,6]. Hypothermia can occur during the summer months and in hospitalized patients [7].

The definition, pathophysiology, and evaluation of accidental hypothermia will be discussed here. A table and algorithm outlining the emergency management of hypothermia in adults are provided (table 1 and algorithm 1). A temperature unit conversion calculator is provided (calculator 1). The following related content is discussed separately:

- Management of accidental hypothermia in adults (see "Accidental hypothermia in adults: Management")
- Drowning, often complicated by hypothermia (see "Drowning (submersion injuries)")
- External injuries from cold (see "Frostbite: Acute care and prevention" and "Nonfreezing cold water (trench foot) and warm water immersion injuries")
- Accidental hypothermia in children (see "Hypothermia in children: Clinical manifestations and diagnosis" and "Hypothermia in children: Management")

DEFINITIONS AND STAGES OF HYPOTHERMIA

Commonly used definition and stages — Hypothermia is defined as a core temperature below 35°C (95°F). The stage of hypothermia, defined by core temperature, has a large impact on both recognition and treatment. The most commonly used definitions found in the literature are as follows [8,9]:

- **Mild hypothermia** – Core temperature 32 to 35°C (90 to 95°F)
- **Moderate hypothermia** – Core temperature 28 to 32°C (82 to 90°F)
- **Severe hypothermia** – Core temperature <28°C (82°F)

In addition, some experts regard a core temperature <24°C (75°F) or <20°C (68°F) as "profound hypothermia" [10,11].

Because the clinical features of hypothermia differ among patients, and because core temperature measurement is imprecise, the recognition of each stage is more important than exact temperature cut-offs.

Prehospital personnel should measure core temperature, preferably using an esophageal probe in the lower third of the esophagus, in an unresponsive patient who is tracheally intubated. (See 'Temperature measurement' below.)

If temperature measurement is not feasible, prehospital personnel should estimate the severity of hypothermia by observing clinical signs. Because the response to hypothermia varies widely among patients, any clinical staging system can only provide a rough estimate of severity (table 2) [12].

Hypothermia scale and clinical findings

Stage	Core temperature	Clinical findings
Cold stressed (but	35 to 37°C (95 to 98.6°F)	Normal mental status with shivering

Hypothermia scale and clinical findings

- **Cold stress (not hypothermia)** – Normal mental status with shivering. Functioning normally. Able to care for self. Estimated core temperature 35 to 37°C (95 to 98.6°F).
- **Mild hypothermia** – Alert, but mental status may be altered. Shivering present. Not functioning normally. Not able to care for self. Estimated core temperature 32 to 35°C (90 to 95°F).
- **Moderate hypothermia** – Decreased level of consciousness. Conscious or unconscious, with or without shivering. Estimated core temperature 28 to 32°C (82 to 90°F).

Table 2 - larger image below

- **Severe/profound hypothermia** – Unconscious. Not shivering. Estimated core temperature <28°C (<82°F).

Other staging systems — Prehospital personnel, especially in Europe, may use the clinical staging scale described by the International Commission for Mountain Emergency Medicine, sometimes referred to as "the Swiss system" (table 3) [10]:

- **Mild (HT I)** – Normal mental status with shivering. Estimated core temperature 32 to 35°C (90 to 95°F).
- **Moderate (HT II)** – Altered mental status without shivering. Estimated core temperature 28 to 32°C (82 to 90°F).
- **Severe (HT III)** – Unconscious. Estimated core temperature 24 to 28°C (75 to 82°F).
- **Severe (HT IV)** – Apparent death. Core temperature 13.7 to 24°C (56.7 to 75°F) (resuscitation may be possible).
- **Death (HT V)** – Death due to irreversible hypothermia. Core temperature <9 to 13.7°C (<48.2 to 56.7°F) (resuscitation not possible).

A limitation of the Swiss staging system is that shivering may not cease until around 30°C (86°F). In addition, vital signs and other signs of life may be present at core temperatures below 24°C (75°F) [13,14].

An analysis of published case reports found a high degree of overlap of core temperatures among the stages and many cases in which the Swiss system overestimated core temperature [15]. The lowest observed core temperatures were 28.1°C (82.6°F) for HT 1, 22.1°C (71.8°F) for HT II, and 19.3°C (66.7°F) for HT III. A further analysis of 305 hospital and published cases found that only 61 percent of patients were assigned to the correct stage [16]. There were large overlaps among the four stages. Core temperature was overestimated in 18 percent and underestimated in 21 percent of patients. The authors calculated that the optimal temperature thresholds to classify patients into the four stages were 32.1, 27.5, and 24°C (89.8, 81.5, and 75°F), which are similar to the original cutoffs of 32, 28, and 24°C (90, 82, and 75°F).

The Swiss system is being replaced by the Revised Swiss System (RSS) as a recommendation of the International Commission for Mountain Emergency Medicine [17]. The RSS stages hypothermia based on the level of consciousness (using the Alert, Verbal, Painful, Unconscious [AVPU] scale) and the presence or absence of vital signs (figure 1). Shivering is no longer a stage-defining sign. The emphasis of the RSS is on the risk of cardiac arrest rather than core temperature. Alert patients (mild hypothermia) have a low risk of cardiac arrest, patients who respond to verbal stimuli have a moderate risk, and patients who respond only to painful stimuli or have no response (unconscious) have a high risk. Patients who are unconscious without detectable vital signs are considered to be in hypothermic cardiac arrest. As with all hypothermia grading systems, the RSS applies only to uncomplicated hypothermia and not to patients with comorbidities such as trauma or sepsis. The risk of hypothermic cardiac arrest would likely be overestimated in patients with comorbidities that might affect the level of consciousness.

A similar system, unrelated to the RSS but only using level of consciousness, has been used in Denmark since 2004. The Danish system has three grades: mild (awake), moderate (unconscious), and severe (lifeless). Severe patients are treated with cardiopulmonary resuscitation (CPR) and extracorporeal membrane oxygenation (ECMO). Retrospective studies have found that level of consciousness alone may be as accurate as core temperature in predicting the risk of cardiac arrest and mortality. In one study of 114 patients, none of the 33 fully alert patients had a cardiac arrest, 3 of the 12 patients who were alert but confused had cardiac arrests, and two-thirds of the 43 patients who were unresponsive had cardiac arrests [18-20].

PATHOPHYSIOLOGY

Body temperature reflects the balance between heat production and heat loss. Heat is generated by cellular metabolism (most prominently in the heart and liver) and lost by the skin and lungs via the following processes [21]:

- Evaporation – Vaporization of water through both insensible losses and sweat
- Radiation – Emission of infrared electromagnetic energy
- Conduction – Direct transfer of heat to an adjacent, cooler object
- Convection – Direct transfer of heat to convective currents of air or water

Of these, convective heat loss to cold air and conductive heat loss to water are the most common mechanisms of accidental hypothermia [22].

The normal set point for human core temperature is 37±0.5°C (98.6±0.9°F) (see "Pathophysiology and treatment of fever in adults", section on 'Normal body temperature'). The human body maintains this temperature whenever possible, using autonomic mechanisms to regulate heat loss and gain in response to environmental conditions. Nevertheless, the human body has limited physiologic capacity to respond to cold environmental conditions. Thus, behavioral adaptations such as clothing and shelter are critical to defend against hypothermia.

The hypothalamus receives input from central and peripheral thermal receptors. In response to a cold stress, the hypothalamus attempts to stimulate heat production through shivering and increased thyroid, catecholamine, and adrenal activity. Sympathetically mediated vasoconstriction minimizes heat loss by reducing blood flow to peripheral tissues, where cooling is greatest [22,23]. Peripheral blood vessels also vasoconstrict in direct response to cold. During removal from the cold environment or rewarming, increased perfusion of cold

extremities occurring from peripheral vasodilation can cause core temperature afterdrop (continued decrease of core temperature) and ventricular arrythmias. (See "Accidental hypothermia in adults: Management", section on 'Prevent core temperature afterdrop/rescue collapse'.)

Cooling decreases tissue metabolism and inhibits neural activity. During the initial phase of cooling, shivering in response to skin cooling produces heat and increases metabolism, ventilation, and cardiac output. Neurologic function begins declining even above a core temperature of 35°C (95°F). Once the core temperature reaches 32°C (90°F), metabolism, ventilation, and cardiac output begin to decline, and shivering becomes less effective until it finally ceases as core temperature continues to drop.

RISK FACTORS

Risk factors associated with death from accidental hypothermia include ethanol or other substance use, experiencing homelessness, psychiatric disease, and older age [2,6,24].

Older adults are at increased risk of developing hypothermia and its complications; causes include decreased physiologic reserve, malnutrition, chronic diseases, medications that impair compensatory responses, and social isolation [24,25]. Mild accidental hypothermia may go unrecognized in older patients as it may mimic other conditions, such as infection, hypothyroidism, or stroke.

Medications can cause or contribute to hypothermia by impairing thermoregulatory mechanisms, decreasing awareness of cold, or clouding judgment. The most common medications that impair thermoregulation are anxiolytics, antidepressants, antipsychotics, and opioids. Medications that can impair compensation for low ambient temperatures include oral antihyperglycemics, beta-blockers, alpha-adrenergic agonists such as clonidine, and general anesthetic agents [26].

In patients with severe hypothermia and spontaneous circulation, bradycardia, hypoxemia, or ventricular arrhythmia is a predictor of developing hypothermic cardiac arrest [27].

CLINICAL MANIFESTATIONS

Clinical manifestations correlate with the degree of hypothermia. As the compensatory mechanisms preventing hypothermia are overwhelmed, the following manifestations typically occur (table 4) [8,28,29]:

Clinical manifestations of accidental hypothermia

Table 4 - larger image below

- **Mild hypothermia** – Manifestations include initial hyperventilation, tachypnea, tachycardia, ataxia, dysarthria, impaired judgment, shivering, and cold diuresis.
- **Moderate hypothermia** – This stage is characterized by approximately linear reductions in heart rate, cardiac output, and respiratory rate as well as central nervous system depression, hyporeflexia, and decreased renal blood flow. Shivering ceases at core temperatures as low as 30°C (86°F). There may be paradoxical undressing. Atrial fibrillation, junctional bradycardia, and other benign arrhythmias can occur. Pupillary light reflexes are depressed with slowing of both constriction and dilation [30]. Dilated pupils are seen below a core temperature of approximately 29°C (84°F).
- **Severe hypothermia** – Manifestations can include coma, areflexia, hypotension, bradycardia, ventricular arrhythmias (including ventricular fibrillation), and asystole [22,31]. Pulmonary edema and oliguria can also occur. Corneal and oculocephalic reflexes can be lost, but this may not correlate with core temperature [32]. Signs that portend a poor prognosis in a normothermic patient, such as fixed and dilated pupils, loss of brainstem reflexes, and apparent rigor mortis are expected and reverse with rewarming if the patient survives.

Vitals signs and level of consciousness should be consistent with the degree of hypothermia. Inconsistency suggests an alternative diagnosis (table 5). (See 'Differential diagnosis' below.)

A retrospective study of 216 hypothermic patients found generally positive linear correlations between core temperature and heart rate, systolic blood pressure, respiratory rate, and Glasgow Coma Scale (GCS) score (table 6) [33]. A calculator to estimate the expected vital signs based on body temperature can be found at the Hypothermia Outcome Prediction after ECLS project at the University Hospital of Lausanne. This tool provides expected vital signs based on hypothermia severity but may not be generalizable to all populations (eg, patients in urban settings) since the study excluded patients who experienced cardiac arrest and those with concomitant diseases or intoxications.

DIAGNOSTIC EVALUATION

Physical examination — In moderate or severe hypothermia, the heart can be very sensitive to movement. Rough handling may precipitate arrhythmias, including ventricular fibrillation. Take care to avoid jostling the patient during removal of clothing, physical examination, or

performance of procedures.

Clinical findings associated with the progressive stages of hypothermia are described above. (See 'Clinical manifestations' above.)

Oxygenation should be monitored continuously. Responses of pulse oximeters placed on fingers are slowed by hypothermia [34]. Topical nitroglycerin may help. Probes placed on the ears or forehead appear to be less influenced by decreased body temperature with associated peripheral vasoconstriction.

A total-body survey should be conducted in all hypothermic patients to exclude local cold-induced injuries in areas not seen during the initial resuscitation and to assess for signs of trauma. Frostbite and local cold-related injury are discussed in detail elsewhere. (See "Frostbite: Acute care and prevention" and "Nonfreezing cold water (trench foot) and warm water immersion injuries".)

Temperature measurement — A digital thermometer should be used whenever possible. In patients who are not tracheally intubated, we prefer a flexible temperature probe placed rectally to measure core temperature and for ongoing monitoring [35,36]. Ensure the digital thermometer is low-reading; depending on the manufacturer, medical-grade digital thermometers with external probes may read as low as 26.7 to 30°C (80 to 86°F). Standard glass thermometers are unsuitable because they read only to a minimum of 34°C (93°F). Infrared tympanic thermometers and temporal artery thermometers are not accurate at any temperature [11].

In patients with severe hypothermia requiring tracheal intubation, an esophageal probe inserted into the lower third of the esophagus (approximately 24 cm below the larynx in adults) provides a close approximation of cardiac temperature [11]. An instructional video demonstrating esophageal temperature measurement is available in the references [37,38]. Rough handling should be avoided so as not to provoke ventricular fibrillation. If heated, humidified oxygen is used, an esophageal probe advanced no farther than the upper two-thirds of the esophagus may read falsely high.

Use of rectal temperature is reasonable in conscious patients, but a rectal probe may provide a falsely low reading if adjacent to cold feces. Bladder temperatures are commonly used and are adequate in mild to moderate hypothermia but should not be used in critical patients to guide rewarming. (See "Accidental hypothermia in adults: Management", section on 'Temperature'.)

Diagnosis — The diagnosis of accidental hypothermia is made based upon a history or other evidence of environmental exposure to cold and a core temperature below 35°C (95°F). Accurate diagnosis depends on the use of a digital or low-reading glass thermometer to determine the core temperature accurately.

ANCILLARY STUDIES IN MODERATE OR SEVERE HYPOTHERMIA

Previously healthy patients with mild accidental hypothermia may not require laboratory investigation.

Laboratory studies — After making the diagnosis of moderate or severe hypothermia, laboratory evaluation should be undertaken to identify potential complications and comorbidities, including lactic acidosis, rhabdomyolysis, bleeding diathesis, and infection, such as the following (table 7) [22,23,28,31,39-41]:

- Fingerstick glucose
- Basic serum electrolytes, including potassium and calcium
- Blood urea nitrogen (BUN) and creatinine
- Hemoglobin, white blood cell (WBC), and platelet counts
- Serum lactate
- Prothrombin time (PT), activated partial thromboplastin time (aPTT), international normalized ratio (INR), and fibrinogen
- Creatine kinase (CK)
- Lipase
- Arterial blood gas (ABG; uncorrected for temperature) in ventilated patients

Additional studies are obtained based on clinical circumstances. A serum ethanol concentration may be helpful in a patient with a depressed mental status that is not commensurate with the core temperature. Serum cortisol, thyroid function studies, and evaluation for infectious sources might be obtained in a patient with moderate or severe hypothermia who did not have an environmental exposure or who fails to rewarm despite aggressive interventions. (See "Accidental hypothermia in adults: Management", section on 'Failure to rewarm'.)

Issues with interpretation of laboratory studies during hypothermia include the following:

- Hematocrit – The hematocrit increases 2 percent for each 1°C (1.8°F) drop in temperature. A low normal hematocrit is abnormal in severe hypothermia [42]. (See "Automated complete blood count (CBC)", section on 'RBC parameters'.)

- Coagulation tests – Hypothermic inhibition of enzymes in the coagulation cascade causes a bleeding diathesis [40]. Because tests of coagulation, such as PT and aPTT are performed at 37°C (98.6°F), the laboratory will report deceptively normal results despite an obvious in vivo coagulopathy. (See "Clinical use of coagulation tests".)

- ABG – Therapy should be guided based on **uncorrected** ABG values (ie, the value reported by modern automated blood gas analyzer at the operating temperature of 37°C [98.6°F]) instead of values corrected for the patient's body temperature [43]. The pH, partial pressure of carbon dioxide ($PaCO_2$), and partial pressure of oxygen (PaO_2) of a blood sample all vary with the temperature; gas tensions and hydrogen ion concentration decline as temperature drops (table 8) [43,44]. An uncorrected pH of 7.4 and a $PaCO_2$ of 40 mmHg reflect normal acid-base balance at 37°C (98.6°F). Maintaining normal uncorrected ABG values during rewarming will produce normal ABG values once the patient rewarms to 37°C (98.6°F). Uncorrected pH values (the alpha-stat strategy) are better prognostic indicators than values corrected for core temperature (the pH-stat strategy) [45]. (See "Arterial blood gases", section on 'Transport and analysis'.)

Electrocardiogram — Hypothermia can cause characteristic electrocardiogram (ECG) changes, including prolongation of intervals and J-point elevation in addition to bradycardia, atrial fibrillation, ventricular arrhythmias (including ventricular fibrillation), and asystole [46]. Slowed impulse conduction through potassium channels results in prolongation of all the ECG intervals, including the RR, PR, QRS, and QT intervals [47].

There may also be elevation of the J point if the ST segment is unaltered, producing a characteristic J (or Osborn) wave that reflects distortion of the earliest phase of membrane repolarization (waveform 1) [48]. The height of the J wave is roughly proportional to the degree of hypothermia [49,50]. These findings are most prominent in precordial leads V2 to V5. Similar findings can be observed in patients with early repolarization, hypercalcemia, and Brugada syndrome. Although suggestive of hypothermia, J waves are not pathognomonic and can be found in other conditions, such as subarachnoid hemorrhage and traumatic brain injury. (See "ECG tutorial: ST- and T-wave changes".)

Software for ECG interpretation may be unable to recognize J waves and may misinterpret them as currents of injury (ischemic changes). J waves have been mistaken for ST elevations, leading to unwarranted cardiac catheterization [51]. (See "Electrocardiogram in the diagnosis of myocardial ischemia and infarction", section on 'Early repolarization'.)

Hypothermia can obscure normal premonitory ECG changes commonly associated with hyperkalemia. (See "Clinical manifestations of hyperkalemia in adults", section on 'Cardiac manifestations'.)

Shivering can cause a rhythmic irregularity of the ECG baseline and, occasionally, the QRS complex (waveform 2).

Imaging studies — Radiologic studies in hypothermia are performed based on clinical circumstances. Ensure that the patient is not jostled while obtaining imaging.

- **Respiratory signs or symptoms** – We obtain a chest radiograph to detect signs of aspiration, pneumonia, lung injury, or pulmonary edema in patients with abnormal respiratory findings, including tachypnea, cough, or crackles on auscultation, or with a submersion injury. (See "Drowning (submersion injuries)", section on 'Initial testing'.)
- **Level of consciousness is not proportional to the degree of hypothermia** – We obtain computed tomography (CT) imaging of the brain to evaluate for structural lesions.
- **Trauma** – Patients with multisystem trauma with associated hypothermia should have appropriate imaging. (See "Initial management of trauma in adults".)
- **Diving injury** – A patient with hypothermia from a submersion injury from diving into shallow water or with signs of focal neurologic deficit warrant cervical spine immobilization and appropriate spine imaging. (See "Cervical spinal column injuries in adults: Evaluation and initial management" and "Suspected cervical spine injury in adults: Choice of imaging".)

DIFFERENTIAL DIAGNOSIS

Non-environmental factors — In addition to hypothermia from environmental exposure, many medical conditions can contribute to hypothermia, including hypothyroidism, adrenal insufficiency, sepsis, neuromuscular disease, malnutrition, thiamine deficiency, and hypoglycemia (table 5) [22,28,35,39,52-54]. Although contributing factors for secondary hypothermia may occur simultaneously with accidental hypothermia, as when alcohol or other substance use leads to prolonged environmental exposure, primary hypothermia can usually be diagnosed on the basis of a history of exposure to cold without other complicating factors. (See "Clinical manifestations of hypothyroidism" and "Clinical manifestations of adrenal insufficiency in adults" and "Overview of water-soluble vitamins", section on 'Deficiency' and "Sepsis syndromes in adults: Epidemiology, definitions, clinical presentation, diagnosis, and prognosis" and "Hypoglycemia in adults without diabetes mellitus: Clinical manifestations, causes, and diagnosis" and "Hypoglycemia in adults with diabetes mellitus" and "Respiratory muscle weakness due to neuromuscular disease: Clinical manifestations and evaluation", section on 'Investigating the underlying disorder'.)

Factors that can contribute to accidental hypothermia in adults

Table 5 - larger image below

In older adults, sepsis can manifest with hypothermia. If there is unexplained hypothermia, a probable source of infection, aspiration, failure to rewarm, or other signs suggestive of sepsis, empiric, broad-spectrum antibiotics should be administered. (See "Evaluation and management of suspected sepsis and septic shock in adults".)

Alcohol use disorder and carbon monoxide poisoning have been implicated in some cases of hypothermia [55]. (See "Wernicke encephalopathy" and "Carbon monoxide poisoning".)

Inconsistent vital signs or mental status suggest a secondary cause — Beware of vital signs or mental status inconsistent with the degree of hypothermia (see 'Clinical manifestations' above). The neurologic manifestations of hypothermia vary widely, but the level of consciousness should be consistent with the core temperature. Any inconsistency is a warning that a secondary cause of hypothermia exists, such as the following:

- Relative tachycardia inconsistent with the core temperature suggests hypoglycemia, hypovolemia, or an overdose. (See "Hypoglycemia in adults without diabetes mellitus: Clinical manifestations, causes, and diagnosis" and "Hypoglycemia in adults with diabetes mellitus" and "Etiology, clinical manifestations, and diagnosis of volume depletion in adults" and "Initial management of the critically ill adult with an unknown overdose".)
- Relative hyperventilation suggests an underlying metabolic acidosis, diabetic ketoacidosis, or salicylate overdose. Carbon dioxide production should be decreased in moderate or severe hypothermia. (See "Approach to the adult with metabolic acidosis" and "Diabetic ketoacidosis and hyperosmolar hyperglycemic state in adults: Clinical features, evaluation, and diagnosis" and "Salicylate (aspirin) poisoning: Clinical manifestations and evaluation".)
- If the level of consciousness is not proportional to the degree of hypothermia, suspect a head injury, central nervous system infection, or overdose. (See "Management of acute moderate and severe traumatic brain injury in adults" and "Acute toxic-metabolic encephalopathy in adults" and "Initial management of the critically ill adult with an unknown overdose".)
- Areflexia or paralysis may be caused by a spinal cord injury rather than by hypothermia. (See "Acute traumatic spinal cord injury".)

SOCIETY GUIDELINE LINKS

Links to society and government-sponsored guidelines from selected countries and regions around the world are provided separately. (See "Society guideline links: Management of environmental emergencies" and "Society guideline links: Hypothermia".)

INFORMATION FOR PATIENTS

UpToDate offers two types of patient education materials, "The Basics" and "Beyond the Basics." The Basics patient education pieces are written in plain language, at the 5th to 6th grade reading level, and they answer the four or five key questions a patient might have about a given condition. These articles are best for patients who want a general overview and who prefer short, easy-to-read materials. Beyond the Basics patient education pieces are longer, more sophisticated, and more detailed. These articles are written at the 10th to 12th grade reading level and are best for patients who want in-depth information and are comfortable with some medical jargon.

Here are the patient education articles that are relevant to this topic. We encourage you to print or e-mail these topics to your patients. (You can also locate patient education articles on a variety of subjects by searching on "patient info" and the keyword(s) of interest.)

- Basics topic (see "Patient education: Hypothermia (The Basics)")

SUMMARY AND RECOMMENDATIONS

- **Stages and clinical manifestations** – Hypothermia is defined as a core temperature below 35°C (95°F). It can be further classified by severity (table 2 and table 4) (see 'Definitions and stages of hypothermia' above and 'Clinical manifestations' above):
  - Mild hypothermia – Core temperature 32 to 35°C (90 to 95°F); manifestations include confusion, tachypnea, tachycardia, and increased shivering.
  - Moderate hypothermia – Core temperature 28 to 32°C (82 to 90°F); findings include lethargy, bradycardia and atrial arrhythmias, decreased or absent pupillary reflexes, and decreased or absent shivering.
  - Severe hypothermia – Core temperature below 28°C (82°F); findings include coma, hypotension, ventricular arrhythmias, including ventricular fibrillation and asystole, pulmonary edema, and loss of corneal and oculocephalic reflexes.

  Vital signs and level of consciousness should be consistent with the degree of hypothermia. There are generally positive linear correlations between core temperature and heart rate, systolic blood pressure, respiratory rate, and Glasgow Coma Scale (GCS) (table 6).
- **Risk factors** – Factors contributing to the development of hypothermia include outdoor exposure, cold water submersion, medical conditions, such as hypothyroidism or sepsis, toxins, such as alcohol or other substances, and medications, such as oral antihyperglycemics and sedative-hypnotics. Risk increases if the ability to autoregulate core temperature is impaired by medications or underlying disease. This is common among older adults. (See 'Risk factors' above.)

- **Evaluation** – In severe hypothermia, the heart is very sensitive to movement. Rough handling may precipitate arrhythmias, including ventricular fibrillation. Take care to avoid jostling the patient during removal of clothing, physical examination, or performance of procedures. Readings of pulse oximeter probes placed on the ear or forehead may be less affected by decreased body temperature compared with readings of those placed on the finger. (See 'Physical examination' above.)

- **Diagnosis** – The diagnosis of accidental hypothermia is made based upon a history or other evidence of environmental exposure to cold and a core temperature below 35°C (95°F). Accurate diagnosis depends on use of a digital or low-reading glass thermometer to measure core temperature. Many standard glass thermometers only read down to a minimum of 34°C (93°F). An esophageal temperature probe is preferred in severe hypothermia. A rectal or bladder temperature is a reasonable option in conscious patients or in mild to moderate hypothermia. (See 'Temperature measurement' above and 'Diagnosis' above.)

- **Laboratory testing** – In a patient with moderate or severe hypothermia, laboratory studies are used to identify potential complications and comorbidities and include the following: fingerstick blood glucose, basic serum electrolytes, including potassium and calcium, blood urea nitrogen and creatinine, lactate, complete blood count, coagulation tests, creatine kinase, lipase, and arterial blood gas (table 7). (See 'Laboratory studies' above.)

- **Electrocardiogram** – Hypothermia causes characteristic electrocardiogram (ECG) changes (interval prolongation, J-point elevation (waveform 1)) in addition to bradycardia, ventricular arrhythmias (including ventricular fibrillation), and asystole. Hypothermia may obscure normal premonitory ECG changes commonly associated with hyperkalemia. Shivering can cause a rhythmic irregularity of the ECG baseline and, occasionally, the QRS complex (waveform 2). (See 'Electrocardiogram' above.)

- **Differential diagnosis** – In addition to hypothermia from environmental exposure, many medical conditions can contribute to hypothermia, including hypothyroidism, adrenal insufficiency, sepsis, neuromuscular disease, malnutrition, thiamine deficiency, and hypoglycemia (table 5). In older adults, sepsis can manifest as hypothermia. Beware of vital signs or mental status inconsistent with the degree of hypothermia as that may be a warning that a secondary cause of hypothermia exists. (See 'Differential diagnosis' above.)

- **Management** – A table and algorithm outlining the management of hypothermia in adults are provided (table 1 and algorithm 1) and discussed separately. (See "Accidental hypothermia in adults: Management".)

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Topic 178 Version 54.0

GRAPHICS

Table 1: **Initial management of hypothermia: Rapid overview of emergency management**

General
Definition: core temperature lower than 35°C (95°F) ▪ Mild: 32 to 35°C (90 to 95°F) ▪ Moderate: 28 to 32°C (82 to 90°F) ▪ Severe: below 28°C (82°F)
In patients who are not completely alert and oriented, measure core temperature with a digital temperature probe, preferably rectally whenever possible. Measure esophageal temperature in patients with tracheal tube or supraglottic airway in place.* Standard oral glass thermometers do not read below 34°C (93°F).
Consider hypothermia secondary to other conditions (eg, infection, hypoglycemia, adrenal insufficiency, hypothyroidism, overdose, trauma). Older adults are at higher risk. Hypothermia with associated comorbidities or trauma is considered more severe.
Clinical aspects
Physical examination ▪ Vital signs • Mild hypothermia: tachypnea, tachycardia, hyperventilation • Moderate hypothermia: expect bradycardia (tachycardia suggests hypoglycemia, hypovolemia, or overdose), hypotension, hypoventilation • Severe hypothermia: hypotension, cardiovascular collapse ▪ Neurological examination • Mild hypothermia: ataxia, dysarthria, impaired judgement; suspect CNS pathology if patient comatose • Moderate hypothermia: CNS depression • Severe hypothermia: areflexia, fixed and dilated pupils, loss of brainstem reflexes, coma
Laboratory evaluation
For patients with moderate or severe hypothermia, studies to obtain include: fingerstick glucose, coagulation studies, CBC, basic electrolytes, BUN and creatinine, serum lactate, electrocardiogram, plain chest radiograph. Additional studies may be needed. ▪ Clinical coagulopathy may be present despite normal measured coagulation times ▪ Increased hematocrit may reflect hemoconcentration ▪ May see low bicarbonate, suggesting anion-gap acidosis; if so, obtain venous or arterial blood gas
Electrocardiogram ▪ Rhythm abnormalities (atrial fibrillation, sinus bradycardia) may be present ▪ Intervals (PR, QRS, and QTc) may be prolonged ▪ Osborn J waves are characteristic of hypothermia (but can occur with other conditions) • Occur at junction of QRS and ST segments, most prominent in V2 to V5 • Distortion of the earliest phase of membrane depolarization • Computer may misinterpret as ischemic injury pattern
Treatment
Endotracheal intubation may be necessary in obtunded or unconscious patients and those with bronchorrhea
Treat hypotension with warmed, isotonic crystalloid (40 to 42°C [104 to 108°F]) initially, vasopressors (norepinephrine preferred) if necessary
Avoid rough movements and activity, which may induce ventricular fibrillation
Rewarming techniques are based on degree of hypothermia ▪ Mild hypothermia • Remove wet clothing, cover with warm blankets, keep room temperature at approximately 28°C (82°F). • Provide active external rewarming, with forced air warming systems if available. Warmed blankets, heating pads, radiant heat sources can also be used. Avoid burning skin. ▪ Moderate hypothermia • Provide active external rewarming as described above. • Give warmed IV fluids (40 to 42°C) and warmed humidified oxygen as adjuncts (these are not primary rewarming methods). • Beware of afterdrop, a drop in core temperature caused by return of cold blood from the extremities to the core circulation. Rewarm trunk first to minimize afterdrop. ▪ Severe hypothermia: active external rewarming and active internal rewarming (active core rewarming) • Perform interventions for moderate hypothermia. • For hemodynamically stable patients, rewarm with endovascular temperature catheter whenever available. • For hemodynamically unstable patients, rewarm with ECMO (preferred approach) or CPB if possible. If ECMO or CPB not available, can perform continuous venovenous rewarming, hemodialysis, continuous arteriovenous rewarming. • If other interventions unavailable, may perform peritoneal or pleural irrigation with warmed, isotonic saline (40 to 42°C).
Treatment of arrhythmias/cardiac arrest ▪ Initiate CPR in all patients with cardiac arrest; do not perform chest compressions if an organized rhythm is present on the cardiac monitor.* ▪ Arrhythmias may persist until patient rewarmed. ▪ Ignore atrial arrhythmias with slow ventricular response.

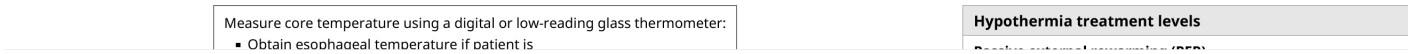
- Ventricular fibrillation is common rhythm – electrical defibrillation may be attempted up to three times but then withheld until core temperature is above 30°C (86°F).
- In a patient with a core temperature <30°C (86°F), do not administer ACLS medications (epinephrine, amiodarone) until the patient is rewarmed to 30°C (86°F), and then renew attempts at defibrillation and resuscitation with ACLS medications. After the patient is rewarmed to 30°C (86°F), double the epinephrine interval as compared with normothermia (every 6 to 10 minutes), and then resume normal intervals when the temperature is ≥35°C (95°F).

BUN: blood urea nitrogen; CBC: complete blood count; CNS: central nervous system; CPB: cardiopulmonary bypass; CPR: cardiopulmonary resuscitation; ECMO: extracorporeal membrane oxygenation; IV: intravenous.

* Refer to UpToDate topic about accidental hypothermia for details about temperature measurement and management of cardiac arrest in severe hypothermia.

Graphic 51697 Version 8.0

Algorithm 1: **Rewarming of adults with accidental hypothermia**



ACLS: advanced cardiac life support; AER: active external rewarming; AIR: active internal rewarming; BLS: basic life support; ECLS: extracorporeal life support; ECMO: extracorporeal membrane oxygenation; IO: intraosseous; IV: intravenous; PER: passive external rewarming.

* We do not advocate chest compressions in patients with an organized, regular cardiac rhythm. In a patient with a core temperature <30°C (86°F), we do not administer ACLS medications (epinephrine, amiodarone) and attempt defibrillation up to three times if they have a ventricular arrhythmia. CPR should continue until the patient is rewarmed to 30°C (86°F), at which point renewed attempts at defibrillation and resuscitation with ACLS medications are undertaken. After the patient is rewarmed to 30°C (86°F), we administer 1 mg IV but with doubled intervals as compared with normothermia (every 6 to 10 minutes), and then resume normal intervals when the temperature is ≥35°C (95°F). Refer to UpToDate topics on accidental hypothermia for details.

¶ Inadequate temperature rise may be due to any number of causes, including sepsis, trauma, intoxication, or stroke. Refer to UpToDate topics on accidental hypothermia for details.

Graphic 131322 Version 4.0

Table 2: **Hypothermia scale and clinical findings**

Stage	Core temperature	Clinical findings
Cold stressed (but not hypothermic)	35 to 37°C (95 to 98.6°F)	Normal mental status with shivering. Functioning normally. Able to care for self.
Mild hypothermia	32 to 35°C (90 to 95°F)	Alert with shivering. Not able to care for self.
Moderate hypothermia	28 to 32°C (82 to 90°F)	Altered level of consciousness. May be conscious or unconscious, with or without shivering.
Severe hypothermia	<28°C (<82°F)	Unconscious. Not shivering.

Graphic 121569 Version 2.0

Table 3: **ICAR-MEDCOM prehospital hypothermia (HT) scale**

International Commission for Mountain Emergency Medicine hypothermia scale		
Hypothermia (HT) stage	**Clinical assessment**	**Expected core temperature, C° (F°)***
HT I	Clear consciousness with shivering	35-32 (95-89.6)
HT II	Impaired consciousness without shivering	32-28 (89.6-82.4)
HT III	Unconsciousness	28-24 (82.4-75.2)
HT IV	Apparent death	<24-13.7 (<75.2-56.7)¶
HT V	Death due to irreversible hypothermia	<13.7-9 (56.7-48.2)¶

* Clinical assessment may be used to stage hypothermia when core temperature measurement in the field is not available, but be aware that clinical findings can vary substantially.
¶ The core temperature at which irreversible hypothermia with death occurs is **not** well defined. Please refer to UpToDate topics on hypothermia for discussions regarding prehospital declaration of death and withdrawal of resuscitation.

Modified with permission from: Durrer B, Brugger H, Syme D, and the International Commission for Mountain Emergency Medicine. The medical on-site treatment of hypothermia: ICAR-MEDCOM recommendation. High Alt Med Biol 2003; 4:99. Copyright © 2003 Mary Ann Liebert, Inc.

Graphic 88073 Version 8.0

Figure 1: **Revised Swiss System for hypothermia and risk of cardiac arrest**

	Stage 1	Stage 2	Stage 3	Stage 4
Clinical findings*	"Alert" from AVPU	"Verbal" from AVPU	"Painful" or "Unconscious" from AVPU **AND** Vital signs present	"Unconscious" from AVPU **AND** No detectable vital signs¶
Risk of cardiac arrestΔ	Low	Moderate	High	Hypothermic cardiac arrest

AVPU: alert, verbal, painful (ie, responds to pain), unconscious; GCS: Glasgow Coma Scale.
* In the Revised Swiss System, "alert" corresponds to a GCS score of 15, "verbal" corresponds to a GCS score of 9 to 14 and includes confused patients, and "painful" and "unconscious" correspond to a GCS score <9. While shivering is not used as a stage-defining sign, its presence usually means that the core temperature is >30°C, a temperature at which hypothermic cardiac arrest is unlikely.
The estimated risk of cardiac arrest is based on accidental hypothermia being the only cause of clinical findings. If other conditions impair consciousness, such as asphyxia, intoxication, high-altitude cerebral edema, or trauma, the Revised Swiss System may falsely predict a higher risk of cardiac arrest. Be cautious with patients who remain "alert" or "verbal" but show signs of hemodynamic or respiratory instability (eg, bradycardia, bradypnea, hypotension); they may be at a transitional stage with a higher risk of cardiac arrest.
¶ No respiration, no palpable carotid or femoral pulse, no measurable blood pressure. Check for signs of life (pulse and, especially, respiration) for up to 1 min.
Δ The transition of colors between stages represents the overlap of patients within groups.

From: Musi ME, Sheets A, Zafren K, et al. Clinical staging of accidental hypothermia: The Revised Swiss System: Recommendation of the International Commission for Mountain Emergency Medicine (ICAR MedCom). Resuscitation 2021; 162:182. Available at: https://www.sciencedirect.com/science/article/pii/S0300957221000964?via%3Dihub (Accessed on November 12, 2021).
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Table 4: **Clinical manifestations of accidental hypothermia**

	Mild	Moderate	Severe
Neurologic	Confusion, slurred speech, impaired judgment, amnesia	Lethargy, hallucinations, loss of pupillary reflexes, EEG abnormalities	Loss of cerebrovascular regulation, decline in EEG activity, coma, loss of ocular reflexes
Cardiovascular	Tachycardia, increased cardiac output and systemic vascular resistance	Progressive bradycardia (unresponsive to atropine); decreased cardiac output and BP; atrial and ventricular arrhythmias, J (Osborn) wave on ECG	Decline in BP and cardiac output, ventricular fibrillation (<28°C; 82.4°F) and asystole (<20°C; 68°F)
Respiratory	Tachypnea, bronchorrhea	Hypoventilation, decreased oxygen consumption and CO2 production, loss of cough reflex	Pulmonary edema, apnea
Renal	Bladder atony, cold diuresis	Cold diuresis	Decreased renal perfusion, oliguria
Musculoskeletal	Increased shivering	Decreased shivering (<32°C; 89.6°F) muscle rigidity	Patient may appear dead ("pseudo-rigor mortis")
Metabolic	Increased metabolic rate, hyperglycemia		Decreased metabolic rate, hyper- or hypoglycemia
Hematologic	Increase in hematocrit, decreased platelet count and white blood cell count, bleeding diathesis, DIC		
Gastrointestinal	Ileus, pancreatitis, gastric stress ulcers, hepatic dysfunction		

BP: blood pressure; CO2: carbon dioxide; DIC: disseminated intravascular coagulation; ECG: electrocardiogram; EEG: electroencephalogram.

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Table 5: **Factors that can contribute to accidental hypothermia in adults**

Environmental exposure
Metabolic/endocrine
▪ Adrenal insufficiency ▪ Hypothyroidism ▪ Hypopituitarism ▪ Hypoglycemia ▪ Diabetes mellitus ▪ Malnutrition ▪ Thiamine deficiency ▪ Anorexia nervosa ▪ Uremia
Iatrogenic
▪ Perioperative hypothermia ▪ Infusion of cold or room temperature IV fluids or blood products ▪ Cardiopulmonary bypass ▪ Kidney replacement therapy with cold dialysate
Infection/inflammatory response
▪ Sepsis ▪ Pancreatitis ▪ Shock
Neurologic/neuromuscular disease
▪ Stroke ▪ Spinal cord injury ▪ Intracranial hemorrhage ▪ Traumatic brain injury ▪ Hypothalamic dysfunction/lesions ▪ Parkinsonism ▪ Multiple sclerosis ▪ Neuropathies
Drugs/toxins
▪ Ethanol ▪ Sedative/hypnotics (eg, benzodiazepines, barbiturates) ▪ Opioids ▪ Alpha-adrenergic agonists (eg, clonidine) ▪ Beta-adrenergic antagonists ▪ Antidepressants (eg, amitriptyline) ▪ Antipsychotics (eg, chlorpromazine) ▪ Carbon monoxide
Others
▪ Trauma ▪ Burns and exfoliative skin disorders (eg, epidermolysis bullosa, exfoliative dermatitis) ▪ Carcinomatosis ▪ Contributing factors (vascular insufficiency, extremes of age, impaired shivering) ▪ Shapiro syndrome

Graphic 58684 Version 3.0

Table 6: **Expected vital signs according to hypothermia stage**

Vital sign*	35°C	32°C	30°C	28°C	26°C	24°C	22°C	20°C
Respiratory rate, breaths/minute (95% CI)	20 (8-32)	18 (6-30)	17 (5-29)	16 (4-28)	15 (3-27)	14 (2-26)	13 (1-25)	12 (0-24)
Heart rate, beats/minute (95% CI)	82 (35-129)	74 (27-121)	69 (22-116)	64 (17-111)	59 (12-106)	54 (7-101)	49 (2-96)	44 (0-91)
Systolic blood pressure, mmHg (95% CI)	139 (86-192)	126 (73-179)	118 (65-171)	109 (52-162)	100 (47-153)	91 (38-144)	83 (30-136)	74 (21-127)
Glasgow Coma Scale (GCS) score (95% CI)	15 (9-15)	13 (7-15)	11 (5-15)	9 (3-15)	8 (3-14)	6 (3-12)	4 (3-10)	3 (3-9)

* Mean vital signs can be used as a guide. Vital signs that fall within the 95% confidence intervals should not necessarily be considered normal for a given core temperature, especially at colder core temperatures.

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Table 7: **Laboratory findings in hypothermia**

Arterial blood gas	Metabolic acidosis, respiratory alkalosis, or both
Electrolytes	No consistent abnormality
Glucose	Increased, decreased, or no change
White blood cell and platelets counts	Decreased due to splenic sequestration
Hemoglobin, hematocrit	Increased due to hemoconcentration
Lipase	May be increased due to hypothermia-induced pancreatitis
Prothrombin and partial thromboplastin times	Increased in vivo due to inhibition of coagulation cascade, despite normal reported values
Electrocardiogram	Prolongation of PR, QRS, QT intervals
	ST segment elevation
	T wave inversions; Osborn J wave
	Atrial fibrillation or sinus bradycardia
Chest radiograph	Aspiration pneumonia, vascular congestion, pulmonary edema

Redrawn from: Mechem CC. Hypothermia and hyperthermia. In: The Intensive Care Unit Manual, Lanken PN, Manaker S, Hanson CW III (Eds), WB Saunders, Philadelphia 2000. p.627.

Graphic 67294 Version 4.0

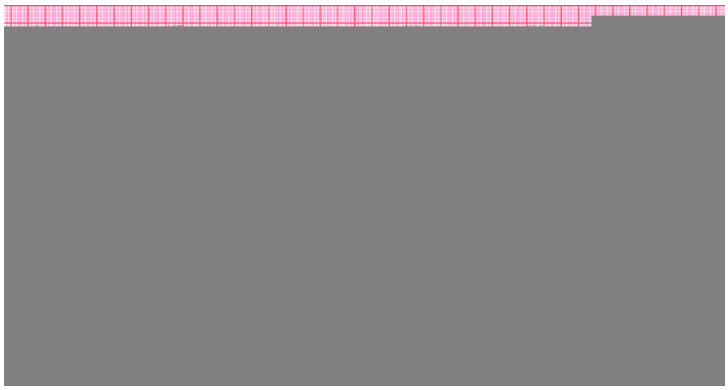
Table 8: **The effect of temperature on blood gas measurements**

Temperature		pH	PCO_2	PO_2
°C	°F			
20	68	7.65	19	27
30	86	7.50	30	51
35	95	7.43	37	70
36	97	7.41	38	75
37	98	7.40	40	80
38	100	7.39	42	85
40	104	7.36	45	97

Adapted from: Shapiro, Peruzzi, Kozelowski-Templin: Clinical Application of Blood Gases, ed 5. St. Louis, Mosby-Year Book, 1994, p. 128.

Graphic 59630 Version 2.0

Waveform 1: **Electrocardiogram in hypothermia**

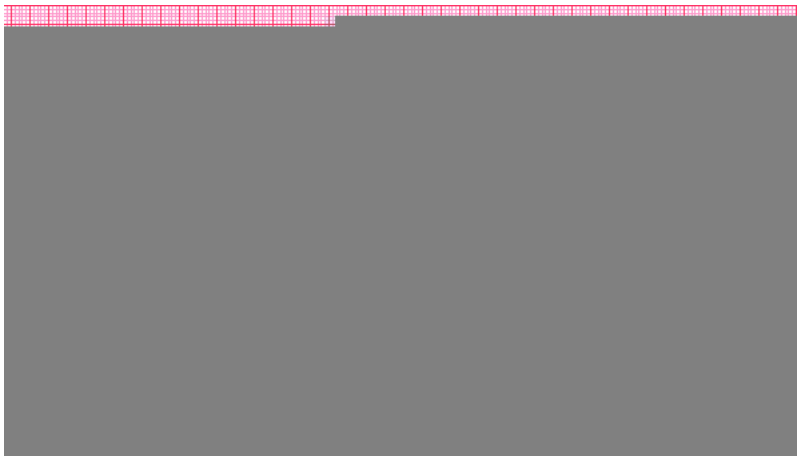


The ECG reveals marked sinus bradycardia (about 40 beats/minute) with first-degree atrioventricular block (PR interval = 0.23 seconds). The slow heart rate in this patient is due to hypothermia (90°F, 32.2°C), which also produces prominent convex deflections at the J point (junction of QRS and ST segments) that are best seen in the precordial leads. The J waves or Osborn waves (arrows) are characteristic of severe hypothermia and resolve with rewarming.

Courtesy of Ary Goldberger, MD.

Graphic 64520 Version 4.0

Normal ECG

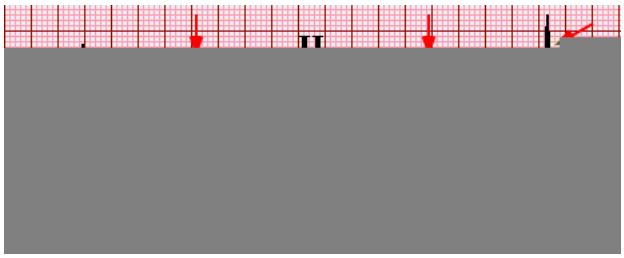


Normal electrocardiogram showing normal sinus rhythm at a rate of 75 beats/minute, a PR interval of 0.14 seconds, a QRS interval of 0.10 seconds, and a QRS axis of approximately 75°.

Courtesy of Ary Goldberger, MD.

Graphic 76183 Version 4.0

Waveform 2: **Tremor artifact**



This dramatic example of tremor artifact demonstrates complexes that simulate a run of ventricular tachycardia. However, QRS complexes (arrows) can clearly be seen marching through the rhythm strip.

Graphic 62575 Version 3.0

Contributor Disclosures

Ken Zafren, MD Consultant/Advisory Boards: SERB Pharmaceuticals [Frostbite]. All of the relevant financial relationships listed have been mitigated. **Daniel F Danzl, MD** No relevant financial relationship(s) with ineligible companies to disclose. **Michael Ganetsky, MD** No relevant financial relationship(s) with ineligible companies to disclose.

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